Procedure:

(i) The straight line trend is represented by the equation $Y = a + bX$ …(1)

where Y is the actual value, X is time, a, b are constants

(ii) The constants 'a' and 'b' are estimated by solving the following two normal Equations

$$\Sigma Y = n\,a + b\,\Sigma X \quad ...(2)$$

$$\Sigma XY = a\,\Sigma X + b\,\Sigma X^2 \quad ...(3)$$

Where 'n' = number of years given in the data.

(iii) By taking the mid-point of the time as the origin, we get $\Sigma X = 0$

(iv) When $\Sigma X = 0$, the two normal equations reduces to

$$\Sigma Y = n\,a + b\,(0);\ a = \frac{\sum Y}{n} = \bar{Y}$$

$$\Sigma XY = a(0) + b\,\Sigma X^2\ ;\ b = \frac{\sum XY}{\sum X^2}$$

The constant 'a' gives the mean of Y and 'b' gives the rate of change (slope).

(v) By substituting the values of 'a' and 'b' in the trend equation (1), we get the Line of Best Fit.



The constant 'a' takes only positive values, but constant 'b' takes both positive and negative values

Example 9.6

Given below are the data relating to the production of sugarcane in a district.

Fit a straight line trend by the method of least squares and tabulate the trend values.

Year	2000	2001	2002	2003	2004	2005	2006
Prod. of Sugarcane	40	45	46	42	47	50	46

Solution:

Computation of trend values by the method of least squares (ODD Years).

Year (x)	Production of Sugarcane (Y)	$X = (x - 2003)$	X^2	XY	Trend values (Yt)
2000	40	–3	9	–120	42.04
2001	45	–2	4	–90	43.07
2002	46	–1	1	–46	44.11
2003	42	0	0	0	45.14
2004	47	1	1	47	46.18
2005	50	2	4	100	47.22
2006	46	3	9	138	48.25
$N = 7$	$\Sigma Y = 316$	$\Sigma X = 0$	$\Sigma X^2 = 28$	$\Sigma XY = 29$	$\Sigma Yt = 316$

Table 9.5

$$a = \frac{\sum Y}{n} = \frac{316}{7} = 45.143$$

$$b = \frac{\sum XY}{\sum X^2} = \frac{29}{28} = 1.036$$

Therefore, the required equation of the straight line trend is given by

$$Y = a + bX$$

$$Y = 45.143 + 1.036\,(x - 2003)$$

The trend values can be obtained by

When $X = 2000$,
$Yt = 45.143 + 1.036\,(2000 - 2003) = 42.035$
When $X = 2001$,
$Yt = 45.143 + 1.036\,(2001 - 2003) = 43.071$,
similarly other values can be obtained.

Example 9.7

Given below are the data relating to the sales of a product in a district.

Fit a straight line trend by the method of least squares and tabulate the trend values.

Year	1995	1996	1997	1998	1999	2000	2001	2002
Sales	6.7	5.3	4.3	6.1	5.6	7.9	5.8	6.1

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